

Table 1: The capability and the accuracy effect across model families, scales and architectures. “Derivations” is the fraction of the generator’s own problems at chain length 25 for which a checker accepts the written proof and the answer is correct. “PW d3” is ProofWriter at inference depth 3. Seed counts are in parentheses and the spread over seeds follows the mean where more than one exists. Every model is a base model given identical instruction tuning, the sparse model has one seed, and the only difference between arms is the content of the replaced five or ten percent of the mixture.

| model        | kind   | derivations, chain 25 |        |         | ProofWriter d3 |        |             |
|--------------|--------|-----------------------|--------|---------|----------------|--------|-------------|
|              |        | control               | formal | English | control        | formal | English     |
| Qwen2.5-3B   | dense  | 0.0                   | 96.0   | 100.0   | 38.2           | 42.1   | 41.8        |
| Qwen2.5-7B   | dense  | 0.3                   | 99.3   | 100.0   | 39.3           | 50.0   | <b>53.2</b> |
| Qwen2.5-14B  | dense  | 7.5                   | 99.0   | –       | 33.0           | 39.2   | 43.6        |
| Llama-3.1-8B | dense  | 0.0                   | 99.5   | 100.0   | 33.3           | 36.3   | 48.8        |
| OLMo-2-7B    | dense  | 0.0                   | 0.5    | 19.5    | 45.8           | 33.6   | 57.2        |
| OLMoE-1B-7B  | sparse | 0.0                   | 0.0    | 1.5     | 59.6           | 33.4   | 26.6        |

Table 2: Two instruction corpora on the same base model install opposite answer priors, and the same replacement corrects both toward the label distribution. The gold labels of ProofWriter at inference depth 3 are 42.8 percent false, 39.8 true and 17.4 unknown. “Excess” is the largest gap between the model’s answer distribution and that distribution. The Tulu treated arm has three seeds, of which one failed to learn the format and is listed separately.

| instruction corpus | condition                     | answer distribution | excess      | PW d3       |
|--------------------|-------------------------------|---------------------|-------------|-------------|
| first corpus       | control                       | true .73            | true +23.5  | 39.3        |
| first corpus       | plus 5% formal                | true .50            | true +0.2   | <b>50.0</b> |
| Tulu 3             | control                       | false .82           | false +39.6 | 53.6        |
| Tulu 3             | plus 5% formal                | false .70           | false +27.0 | <b>57.2</b> |
| Tulu 3             | third seed, format not learnt | as control          | –           | 51.0        |

Table 3: The depth of the training proofs decides how far the procedure extends, and the rendering decides how far it extrapolates. Entries are the fraction of the generator’s own problems whose written proof a checker accepts, two seeds per cell, mean over seeds. Benchmark accuracy over the same runs sits between 46.4 and 51.0 on ProofWriter at depth 3 and between 54.2 and 58.1 on FOLIO, against a control of 39.3 and 51.4, with no dependence on the training band or the rendering.

| training band | notation | chain 5 | chain 15 | chain 25    |
|---------------|----------|---------|----------|-------------|
| 5             | formal   | 99.8    | 13.3     | 11.8        |
| 5             | English  | 100.0   | 85.5     | 55.0        |
| 15            | formal   | 100.0   | 100.0    | 45.0        |
| 15            | English  | 100.0   | 99.8     | <b>99.5</b> |
| 25            | formal   | 99.8    | 100.0    | 99.3        |